

FACULTY OF ECONOMIC AND MANAGEMENT SCIENCES

DEPARTMENT OF ECONOMICS

ECONOMICS 114 A1

DATE:	12 April 2024
DURATION:	2 hours
TOTAL OF MARKS:	60
EXAMINERS:	Dr C Links (convenor) Prof S du Plessis Dr D Shepherd Dr K Malindi Dr P Vorster Mr J Pretorius Mr M Meyer Mrs O Ezeobi
INTERNAL MODERATOR:	Dr E Moses

PLEASE NOTE: ONLY NON-PROGRAMMABLE CALCULATORS ARE ALLOWED.

ANSWER **SECTION A**'s QUESTIONS ON THE SUPPLIED **ANSWER SHEET**. INDICATE YOUR CHOICE BY MAKING A **CROSS** NEXT TO THE NUMBER OF THE STATEMENT.

ALSO ANSWER **SECTION B**'s QUESTIONS ON THE SUPPLIED **ANSWER SHEET**. INDICATE YOUR CHOICE BY MAKING A **CROSS** NEXT TO THE NUMBER OF THE STATEMENT.

ANSWER **SECTION C**'s QUESTIONS IN THE ANSWER BOOK PROVIDED. START EVERY QUESTION ON A NEW PAGE. SHOW ALL CALCULATIONS

KEY:

AP = Average product

BL = Budget line

GDP = Gross domestic product

IC = indifference curve

K = capital

L = labour

FF = Feasible frontier

MP = Marginal product

MRS = Marginal rate of substitution

MRT = Marginal rate of transformation

P = Price

PF = Production function

Q = Quantity

TP = Total product

w = Wage

Y = Income

SECTION A

TRUE OR FALSE STATEMENTS

[6 marks: 1 mark for a correct answer and - ½ for an incorrect answer]

1. Amazon Web Services (AWS) was established in Cape Town, South Africa, nearly two decades ago, and operates as a subsidiary of the American tech giant, Amazon. The revenue generated by AWS in South Africa is included in the calculation of South Africa's GDP.
a) True
b) False
2. According to Malthusian Theory, income growth leads to population growth, which then results in further income growth as more labour becomes available.
a) True
b) False
3. The MRT is the slope of a consumer's indifference curve.
a) True
b) False
4. The MRS is the rate at which a consumer is willing to give up one good to consume an additional unit of another good whilst moving onto a higher indifference curve.
a) True
b) False
5. The Nash equilibrium of the game shown below is (Don't vote, Don't vote).

		Player 2	
		Vote	Don't vote
Player 1	Vote	5, 5	2, 6
	Don't' vote	6, 2	3, 3

- a) True
b) False
6. Property owners wield bargaining power when employing strategic measures like the Proposer in an Ultimatum game.
a) True
b) False

SECTION B
MULTIPLE CHOICE QUESTIONS

[24 marks: 2 marks for a correct answer and – ¼ for an incorrect answer]

7. The Simoleon (§) is the national currency of SimNation. By referring to the table below, which ONE of the following statements is CORRECT?

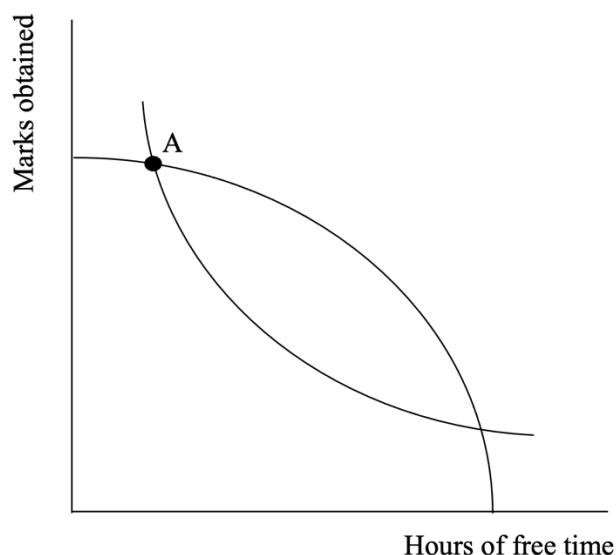
YEAR	2022	2023
GDP	§3 419 million	§3 334 million
POPULATION	175.3 million	170.6 million

- a) SimNation's GDP declined by 2.53% from 2022 to 2023.
 - b) SimNation's GDP increased by 2.49% from 2022 to 2023.
 - c) the fall in SimNation's population offset the fall in GDP so that there was overall growth in GDP per capita between 2022 and 2023.
 - d) the fall in SimNation's population together with the fall in GDP resulted in a decline in GDP per capita between 2022 and 2023.
8. If a cup of coffee costs R30 and a slice of cake costs R50, then
- a) the relative price of coffee is R20.
 - b) the relative price of coffee is 0.6 slices of cake.
 - c) the relative price of coffee is 1.67 slices of cake.
 - d) coffee is too expensive compared to a slice of cake.
9. Which of the following statements is/are TRUE?
- I. Economic rent is also called a surplus in economics.
 - II. Innovation rent creates an economic incentive for firms.
 - III. Rational decision makers want to earn economic rent.
 - IV. Reservation options are always valued at zero.
- a) I only
 - b) I and II
 - c) I, II, and III
 - d) I, II, III and IV

10. Consider the following production function of Sebastian who grows tomatoes and indicate which statement is CORRECT:

Hours (h) per week	0	12	24	36	48	60	72
Tomatoes (kg) per week	0	16	35	55	71	81	90

- a) Sebastian is most efficient when he works 72 hours a week.
 - b) Sebastian's production function displays a constant average product.
 - c) The marginal product of the 24th hour worked is 0.8 kg of tomatoes.
 - d) Sebastian's average product of hours worked declines past 36 hours of work a week.
11. Consider a consumer with a fixed budget that chooses to consume a combination of two normal goods, X and Y. The consumer's budget line is plotted with good X on the horizontal axis. Which ONE of the following statements is CORRECT?
- a) An increase in the price of good X will cause the budget line to shift to the right, *ceteris paribus*.
 - b) If the price of good X increases, the consumer's real income in terms of good X will decrease, *ceteris paribus*.
 - c) If the price of good X increases, the income effect on good Y will be positive, *ceteris paribus*.
 - d) All of the above are correct.
12. The figure below shows the feasible frontier and an indifference curve of a high school student who faces a trade-off between free time and school marks obtained.



At point A the student is not in equilibrium because _____. He could increase his utility by _____.

- a) $MRT > MRS$; increasing his free time
- b) $MRT > MRS$; studying more
- c) $MRT < MRS$; studying more
- d) $MRT < MRS$; increasing his free time

13. Which statements are TRUE about a consumer's best affordable (optimal) point?
- I. It is situated on the lowest attainable indifference curve but on the budget line.
 - II. It has an MRS between two goods equal to the relative price of the good plotted on the vertical axis.
 - III. It exhausts all available income.
- a) I and III
 - b) I only
 - c) II only
 - d) II and III

14. A worker faces a budget constraint represented by:

$$c = 150(24-t)$$

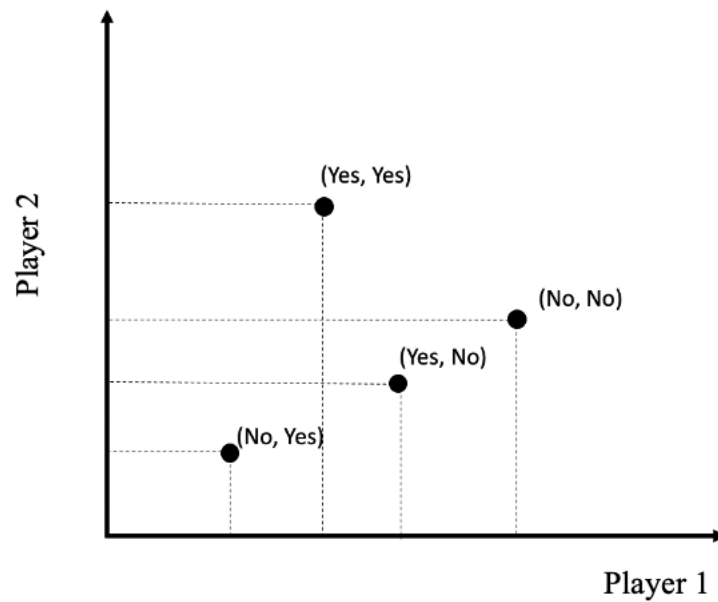
where t is free time and c is consumption, and both are normal goods. Her earnings are paid in South African Rand. Which of the following statements is INCORRECT?

- a) The opportunity cost of an hour of free time for this worker is R150.
 - b) She has R150 worth of unearned income.
 - c) The income effect of a wage increase on her hours of work would be negative.
 - d) The substitution effect of a wage increase on her consumption would be positive.
15. Thandi and Johan get arrested by the police as suspects in a crime. They must decide to confess or not to confess. The number of years they will spend in jail, given the strategy that they choose, is given in the table below. Assume that they are playing a simultaneous, one-shot game with no cooperation possible. Which of the following statements is TRUE?

		Johan	
		Confess	Do not confess
Thandi	Confess	4, 4	1, 8
	Do not confess	8, 1	2, 2

- a) Neither player has a dominant strategy.
- b) Both players can do better if both do not follow their dominant strategy.
- c) The outcome (*confess, confess*) is the optimal outcome for the players of this game.
- d) The Nash equilibrium of this game is Pareto efficient.

Consider the following two-player game represented in graphical form. Use the information contained in this figure to answer question 16 and question 17.



16. Which of the following statements regarding the game is TRUE?

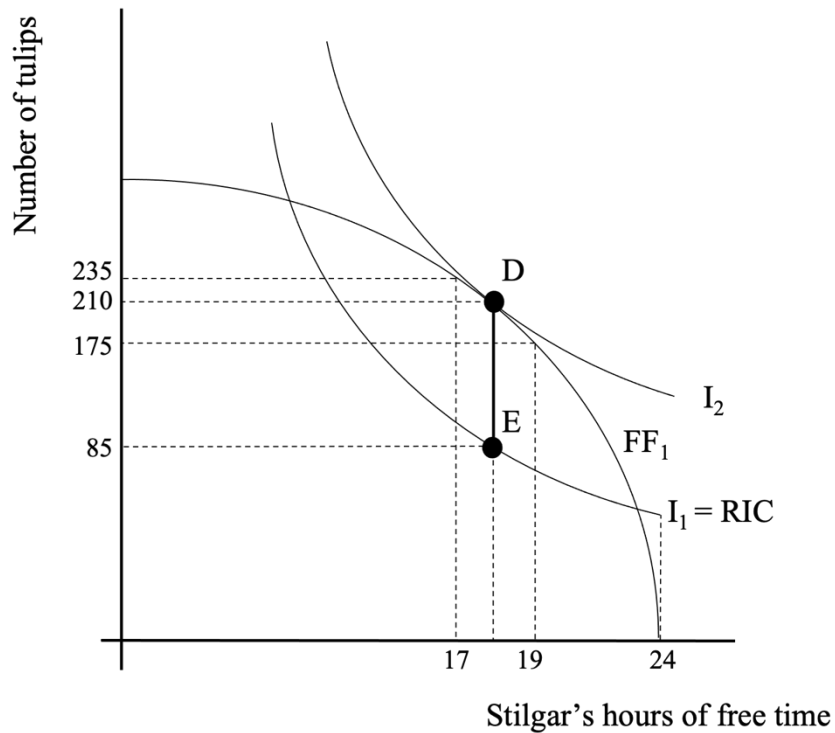
- a) Player 1 has a dominant strategy of "No".
- b) This game has a dominant strategy equilibrium.
- c) This game has two Nash equilibria.
- d) The Nash equilibrium of this game is (Yes, No).

17. Which of the following statements regarding the game is/are TRUE?

- I. The outcome (Yes, Yes) is Pareto inefficient.
- II. There are more than one Pareto efficient outcomes.
- III. (Yes, Yes) is a better outcome than (No, No).

- a) I only
- b) II only
- c) I and II
- d) I, II and III

18. Stilgar works for Vladimir on his farm, producing tulips. Stilgar pays Vladimir part of the harvest as part of a sharecropping rent agreement. The combined feasible frontier (FF_1) and Stilgar's preferences, including his reservation indifference curve (RIC), are represented in the figure below. Which one of the following statements is CORRECT?



- a) A Pareto improvement is possible when the allocation of the production moves from E to D.
- b) E represents a position where Stilgar has more bargaining power than what is represented at D.
- c) At point D the opportunity cost of free time is 60 tulips.
- d) The midpoint on the DE line is the most Pareto efficient allocation.

SECTION C
DESCRIPTIVE QUESTIONS
[30 marks]

QUESTION 19 [3 marks]

In the Star Wars universe, the Empire and the Rebel Alliance are at war with each other. They fight using starships and gathering intelligence. The Wookies and the Ewoks are part of the Rebel Alliance. If the Wookies use all their resources to build starships, they can produce 45 starships in a year. If the Ewoks use all their resources on building starships, they can amass a fleet of 20 starships in a year. If the Wookies use all their resources on gathering intelligence on the Empire, they can gather 15 intelligence reports in a year. The Ewoks can gather 18 intelligence reports in a year if they use all their resources on gathering intelligence.

- 19.1 Who has an absolute advantage in building starships, and who has an absolute advantage in gathering intelligence? [1]
- 19.2 According to the principle of comparative advantage, which activity should the Wookies and the Ewoks prioritise? Show your calculations. [2]

QUESTION 20 [7 marks]

A gold mining company, "Gold Deep", can choose between five different technologies (A, B, C, D, and E) to extract 1 000 000 tonnes of ore. Each technology uses some combination of two inputs, workers and fuel for machinery, as indicated in the table below.

Technology	Number of workers	Fuel for machinery
A	1	6
B	2	7
C	4	4
D	8	8
E	9	1

- 20.1 Each worker is paid R130, and the price of a unit of fuel is R200. Which of the technologies are dominated? [1]
- 20.2 Which technology will the mining company choose to use? Show your calculations. [2]
- The price of a unit of fuel increases to R220, whilst the price of a worker remains unchanged.
- 20.3 Will "Gold Deep" continue to use the technology you identified in (b)? Explain. [2]

Assume a new technology becomes available to "Gold Deep" that allows the same output of 1 000 000 tonnes of ore to be extracted using 3 units of fuel and 2 workers. "Gold Deep" is the only mine with access to this technology. The price of a unit of fuel remains R220 and the price of a worker R130.

20.4 Will "Gold Deep" adopt the new technology? Justify your answer by calculating the innovation rent. [2]

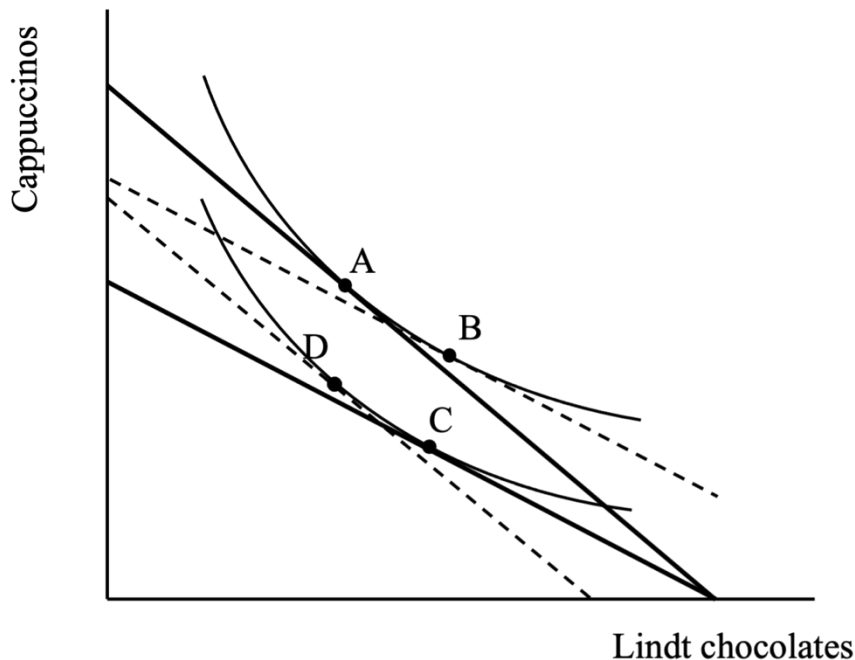
QUESTION 21 [5 marks]

Suppose you share a flat with two other people, Ashley and Lesley, who both have part-time jobs at Gen Z Entertainment. The owner of Gen Z Entertainment increases wages for all staff. Ashley decides to work fewer hours per week from now on, while Lesley decides to work more hours per week. Explain their decisions in terms of the income effect and the substitution effect. Your explanation should include definitions of these effects, as well as how these effects led to Ashley and Lesley's respective decisions.

QUESTION 22 [3 marks]

In the graph below, consumption bundle C represents Daniel's best affordable (optimal) consumption bundle of two goods, Lindt chocolates and cappuccinos. The price of cappuccinos falls. With the aid of the graph, and referring to points A, B, C and D, identify the:

- 22.1 overall price effect
- 22.2 income effect, and
- 22.3 substitution effect



QUESTION 23 [7 marks]

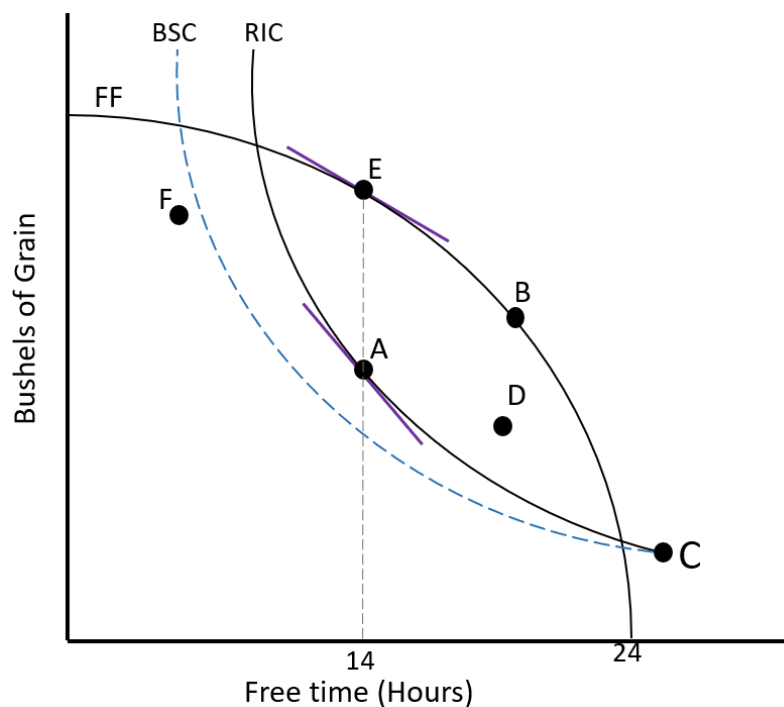
It is the 2024 Indian Premier League Cricket tournament, and the Delhi Capitals are set to face the Chennai Super Kings in the final. Assume that the two teams must choose whether to play tactically or aggressively, and their interaction is represented in the table below.

		Chennai Super Kings	
		Tactical	Aggressive
Delhi Capitals	Tactical	15, 15	5, 23
	Aggressive	23, 5	8, 8

- 23.1 What is the equilibrium outcome of the game? [1]
- 23.2 Is the Nash equilibrium of this game a dominant strategy equilibrium? Motivate your answer fully. [3]
- 23.4 The umpire of the match wants to introduce a fine for any teams that play aggressively. How large would the fine need to be to ensure that (Tactical, Tactical) is the only Nash equilibrium in this game? [2]

QUESTION 24 [6 marks]

Karabo is a farmer who pays a share of her grain output to Dipuo for the use of his land. This is a voluntary exchange, where Dipuo makes Karabo a take-it-or-leave-it offer. The figure below shows Karabo and Dipuo's combined feasible frontier (FF), as well as Karabo's biological survival constraint (BSC) and reservation indifference curve (RIC).



24.1 Explain why Karabo's biological survival constraint curve gets steeper when she works more. [1]

24.2 If Karabo works for 10 hours, would this maximise the economic surplus that can be shared between them? Briefly explain your answer. [2]

A persistent drought that leads to chronic water scarcity results in a permanent reduction in Karabo's grain production.

24.3 Explain what happens to the graph as a result of the drought. [1]

24.4 Which of Karabo and Dipuo would prefer allocation D over allocation A? Justify your answer. [2]